A Neuroscience-Grounded Signature for Belief Imparting by Content: Convergent but Modest Evidence across Text Persuasion, Real fMRI, and an In-Silico Encoder

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Abstract

Large language models can shift human attitudes, and human neuroscience shows that belief change and value-based choice have predictable neural signatures. These literatures have not been joined. We ask whether one neuroscience-grounded signature, computed from content, predicts real belief and choice, generalizes across domains, holds on real fMRI, and can serve as an in-silico instrument that scores content for belief-likelihood. We define the signature as B = engagement + value − resistance and test it in five stages on public data. A synthetic control validates the apparatus. On ChangeMyView (3866 matched argument pairs), the engagement channel is higher for the argument that changed a view (Wilcoxon $p = 9\text{e}{-25}$), though the composite adds no pooled value over content features. On Persuasion-for-Good, the effect points the same way and is significant when pooled (coefficient 0.075, 95% CI [0.036, 0.114]). On real fMRI, the signature computed from brain activity predicts aggregate choice out-of-sample ($r = 0.182$, permutation $p = 0.003$), and a parsimonious approach-minus-avoidance composite generalizes where richer models overfit. An in-silico encoder generalizes across domains and isolates two robust levers (more evidence, less identity threat). The effects are consistent and out-of-sample but modest. The contribution is a validated, interpretable belief-scoring instrument, framed as a measurement and risk-characterization tool.

1 Introduction

Conversational AI can change human attitudes. Model-written messages shift political attitudes about as much as human-written ones (Bai et al., 2025), an effect driven mostly by post-training and prompting rather than personalization or scale (Hackenburg et al., 2025; Argyle et al., 2025), and a meta-analysis finds no mean difference between model and human persuasion but large heterogeneity, with study moderators explaining most of the variance (Hölbling et al., 2025). This literature measures behavioral outcome and carries no neural signal.

Human neuroscience supplies the missing half. Neural responses to persuasive messages predict later behavior change beyond stated intention, with medial prefrontal cortex carrying the signal (Falk et al., 2010). Nucleus accumbens activity forecasts aggregate choice, in several settings better than choice itself (Genevsky et al., 2017; Knutson and Genevsky, 2018; Tong et al., 2020). Belief resistance appears in default-mode and dorsomedial prefrontal cortex (Kaplan et al., 2016). Each is a passive observation, and none asks whether such a signature, read from content, carries information that content analysis misses.

We join the two. The research question is whether one neuroscience-grounded signature, computed from content, predicts real belief and choice, generalizes across domains, holds on real neural data, and can serve as an in-silico belief-scoring instrument. We answer it in five stages on public data. The aim is a validated measurement instrument, not a deployed persuasion system.

2 The Signature

For a piece of content we read three quantities, each tied to a published result: engagement E (medial prefrontal cortex; Falk et al., 2010), value V (nucleus accumbens; Genevsky et al., 2017), and resistance R (default-mode and anterior insula; Kaplan et al., 2016). The signature is $B = z(E) + z(V) - z(R)$. In the text arms E, V, R are computed from transparent lexical channels; in the neural arm they are read from real regional activity; in the in-silico encoder they are intermediate variables constrained to the same structure.

3 Data and Methods

All data are public. ChangeMyView / Winning Arguments (Tan et al., 2016), via ConvoKit: 19571 labeled challenger arguments in 3866 matched pairs, outcome a delta awarded (view changed). Persuasion-for-Good (Wang et al., 2019): 1017 charity-donation dialogues, outcome the persuadee donated. A stock neuroforecasting task (Kuhnen and Knutson, 2005): group-averaged BOLD in nucleus accumbens, medial prefrontal cortex, and anterior insula, with aggregate buy fraction as outcome across two experiments.

Synthetic validation uses a stand-in with a known ground truth and three conditions (signal present, content-only, scrambled). Text arms use within-pair sign tests, grouped cross-validation, bootstrap intervals, per-channel decomposition, and a fixed-effect cross-corpus combination. The neural arm uses cross-experiment out-of-sample prediction with permutation nulls and a six-model comparison. The in-silico encoder is evaluated by cross-domain training, bootstrap factor variance, and counterfactual optimization. All code and results will be released.

4 Results

Apparatus validity. On the synthetic stand-in the instrument detects a planted signal (delta-AUC $+0.067$), reports null when belief is content-only, and returns chance when the signature is scrambled (AUC 0.497). Across twelve seeds the increment test detects the signal in 12 of 12 positive-control seeds and reports the correct null in 11 of 12 content-only seeds.

Real belief change. On ChangeMyView, within matched pairs holding topic and audience fixed, the engagement channel is higher for the argument that changed a view in 57.9% of pairs (Wilcoxon $p = 9\text{e}{-25}$) against a permutation null of 0.506. The value channel is null (0.487, $p = 0.07$); the resistance text-proxy runs opposite to theory, so the composite is dominated by engagement. The signature adds no pooled value over content features (content AUC 0.569, content plus B 0.569), a null we read as consistent with the meta-analytic heterogeneity.

Cross-domain generalization. On Persuasion-for-Good, a different domain, the engagement coefficient points the same way (ChangeMyView $+0.076$, significant; Persuasion-for-Good $+0.059$, same sign, not individually significant), and the pooled estimate is significant ($+0.075$, 95% CI [0.036, 0.114]). Value does not generalize.

Real neural data. On real fMRI the signature predicts aggregate choice out-of-sample (cross-experiment $r = 0.182$, AUC 0.571, permutation $p = 0.003$). The composite is the only predictor sign-stable across experiments; single regions flip sign. The effect is near-null in one experiment ($r = 0.05$) and strong in the other ($r = 0.32$).

What generalizes best. A six-model comparison, guided by the neuroforecasting and small-sample regularization literatures (Genevsky et al., 2017; Tong et al., 2020; Mortazavi et al., 2025), shows the parsimonious approach-minus-avoidance composite (nucleus accumbens minus anterior insula) is the best generalizer ($r = 0.180$, permutation $p = 0.002$). Every multivariate or learned model generalizes worse (ridge on all features $r = 0.093$, not significant). The affect composite is positive across four anticipatory time windows.

In-silico encoder. A brain-structured encoder maps content factors through an approach-minus-avoidance layer to a belief score (Wang et al., 2025; Walters et al., 2022; Kriegeskorte and Douglas, 2018). Trained on one text corpus and applied to the other, the brain-state read-out generalizes slightly better than the unconstrained model both ways (AUC 0.546 and 0.536 versus 0.537 and 0.519). Of six factors, only two are robust across both corpora: evidence raises belief and identity threat lowers it. Counterfactually altering these two on low-belief content raises the predicted belief score (ChangeMyView 0.600 to 0.625, Persuasion-for-Good 0.450 to 0.541).

5 A Formal Model

A companion model places the signature inside a dynamical account: belief is a state in a cusp potential (van der Maas et al., 2003), the persuasive force carries the signature term, gated by bounded confidence (Hegselmann and Krause, 2002) and asymmetric updating (Palminteri and Lebreton, 2022; Lefebvre et al., 2017). Belief robustness is a computable functional (well stiffness and distance to the fold, scaling as involvement to the three-halves). Four predictions are reproduced in the implementation: hysteresis, a foot-in-the-door advantage of sequential in-band messaging, super-linear robustness scaling, and susceptibility divergence at the fold. These are verified in code, not yet against human behavior.

6 Discussion

Three independent analyses converge on the same two levers. The ChangeMyView within-pair test, the real fMRI composite, and the in-silico encoder all point to an approach-and-engagement signal minus an avoidance-and-resistance signal. That the same structure carries a real, out-of-sample, permutation- validated signal across text persuasion and real neural choice, and isolates the same interpretable levers in an in-silico encoder, is the central finding. It is convergent but modest: an eight-point within-pair lift, a cross-experiment r near 0.18, a cross-domain AUC near 0.54. The signature is best used to score content and characterize risk, not to forecast individual outcomes.

Limitations

The text arms use lexical proxies for the neural quantity, not a learned brain encoder. The neural arm uses real fMRI but a different behavior (financial choice) than belief change, and its effect is experiment-dependent. The resistance channel runs opposite to theory in text. No single dataset closes the loop from content to a neural state to a belief outcome in the same people, which does not yet exist publicly at scale. The formal model's predictions are verified in code, not against human data. These bound the claim to convergent, modest, out-of-sample evidence.

Ethics Statement

This work develops and releases a scoring and risk-characterization instrument, not a deployed persuasion system, and treats the optimization as a counterfactual sensitivity analysis rather than a generator pointed at people. All data are public; no new human-subjects data are collected. The interpretable belief levers support detection and inoculation research.

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